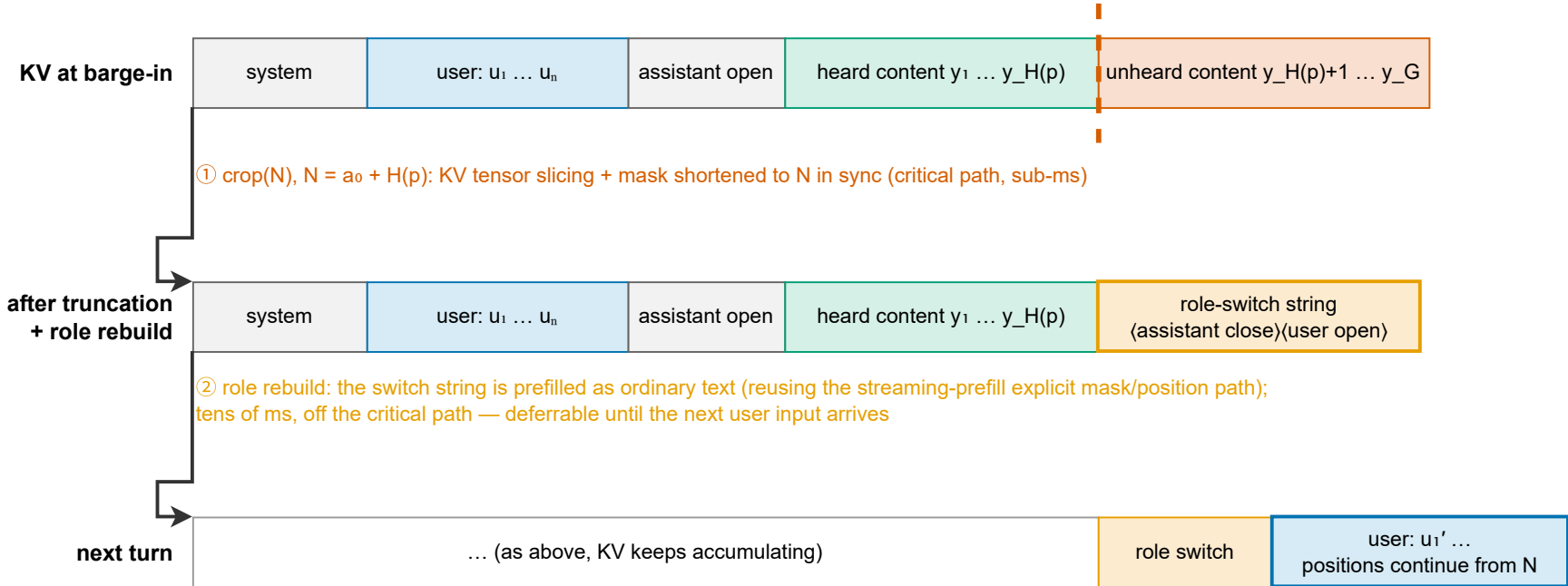




All steps preserve the three invariants: $|\mathcal{K}| = |\text{mask}| = \text{sequence length}$; contiguous position encoding; template-legal roles (runtime-checked in all experiments)